

Virtual Retransmission via Flow Matching for Reliable DeepJSCC Image Transmission

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Abstract—Hybrid automatic repeat request (HARQ) improves neural joint source–channel coding reliability at the cost of another channel round. This work asks whether receiver computation can avoid some of those transmissions. We propose FlowHARQ, a receiver-side virtual retransmission stage that detects unreliable DeepJSCC latent tokens and transports only those tokens toward the clean latent distribution using reliability-anchored flow matching (RAFM). Reliable tokens are invariant by construction. A calibrated post-flow quality head requests a physical second round only when repair remains insufficient; when requested, both rounds are combined using instantaneous-CSI maximum-ratio combining. On a held-out DIV2K protocol with TBD training seeds, TBD channel seeds, and TBD test images, FlowHARQ changes the physical retransmission rate by TBD percentage points (95% CI [TBD, TBD]) relative to adaptive HARQ, while changing PSNR by TBD dB (95% CI [TBD, TBD]). These results test the central premise that flow matching can act as a zero-airtime HARQ round rather than merely a generative image decoder.

I. INTRODUCTION

Deep joint source–channel coding (DeepJSCC) maps images directly to channel symbols and degrades gracefully with SNR [1]. Transformer backbones further improve high-resolution transmission and channel adaptability [2]. Reliability remains difficult in severe fading: always retransmitting wastes airtime, whereas an erroneous ACK produces a poor reconstruction. Recent semantic HARQ methods learn when or what to retransmit [3], [4], but recovery still consumes another wireless round.

Flow matching (FM) learns a continuous vector field without simulating an ODE during training [5]. Existing FM communication receivers use FM for global reconstruction or refinement [6], [7]. Their goal is better or faster decoding, including channel-aware latent transport [8], not deciding whether a physical retransmission can be skipped. Conversely, generative semantic HARQ has been explored for text through stochastic VAE latents [9], leaving image DeepJSCC and FM unconnected at the HARQ decision point.

We introduce *virtual retransmission*: the receiver first repairs only unreliable semantic latents with FM and then issues ACK/NACK. The contribution is one system-level idea with three parts: (i) receiver-side FM before HARQ, (ii) reliability-anchored partial transport that exactly preserves trusted latents, and (iii) a calibrated post-flow quality decision evaluated against adaptive and oracle HARQ under paired, time-correlated fading. The intended operating point is matched reconstruction quality with fewer physical rounds.

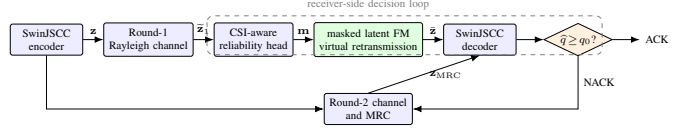


Fig. 1. FlowHARQ inserts a receiver-only virtual retransmission between the first reception and the physical HARQ decision.

II. SYSTEM MODEL

For image $\mathbf{x}$, a SwinJSCC encoder emits N real-valued latent tokens $\mathbf{Z} = E_\theta(\mathbf{x}) \in \mathbb{R}^{N \times d}$. Adjacent latent dimensions form complex symbols. In HARQ round $r \in \{1, 2\}$,

$$\mathbf{y}_r = h_r \mathbf{s} + \mathbf{n}_r, \quad \tilde{\mathbf{Z}}_r = \mathcal{Q}^{-1}(h_r^* \mathbf{y}_r / |h_r|^2), \quad (1)$$

where $\mathcal{Q}^{-1}$ restores token scale and the receiver knows h_r . Fading between rounds follows

$$h_2 = \rho_\Delta h_1 + \sqrt{1 - \rho_\Delta^2} e, \quad \text{quad} \rho_\Delta = J_0(2\pi f_D \Delta), \quad (2)$$

with $e \sim \mathcal{CN}(0, 1)$ and HARQ interval Δ . Thus speed changes the joint, not merely marginal, round statistics. A requested second round is combined using instantaneous-CSI MRC,

$$\mathbf{Z}_{\text{MRC}} = \frac{|h_1|^2 \tilde{\mathbf{Z}}_1 + |h_2|^2 \tilde{\mathbf{Z}}_2}{|h_1|^2 + |h_2|^2}. \quad (3)$$

III. FLOWHARQ

A. Reliability-Anchored Partial Flow

A reliability head outputs token corruption probabilities $q_i = g_q(\tilde{\mathbf{z}}_{1,i}, \mathbf{c}_{\text{rx}})$, where $\mathbf{c}_{\text{rx}}$ includes nominal SNR, speed, CSI age, instantaneous gain, and effective SNR. At inference, $m_i = \mathbb{I}[q_i \geq \tau_r]$. During training, the target mask is the top- κ token-error subset computed from the clean latent.

For masked tokens, we use the straight conditional path

$$\mathbf{z}_i(t) = \tilde{\mathbf{z}}_{1,i} + t m_i (\mathbf{z}_i - \tilde{\mathbf{z}}_{1,i}), \quad \mathbf{u}_i = \mathbf{z}_i - \tilde{\mathbf{z}}_{1,i}. \quad (4)$$

The velocity field is conditioned on the received tokens, receiver CSI, mask, and a pooled anchor computed only from reliable tokens. Its objective is

$$\mathcal{L}_{\text{FM}} = \mathbb{E}_t \frac{\sum_i m_i \|v_\psi(\mathbf{z}_i(t), t, \mathbf{c}_{\text{rx}}, \mathbf{Z}_{\text{rel}}) - \mathbf{u}_i\|_2^2}{\sum_i m_i}. \quad (5)$$

Midpoint Euler integration uses K evaluations. A final projection enforces

$$\bar{\mathbf{z}}_i = (1 - m_i) \tilde{\mathbf{z}}_{1,i} + m_i \mathbf{z}_i^{\text{FM}}, \quad (6)$$

so FM cannot alter a token declared reliable.

Held-out quality–retransmission figure generated after evaluation.

Fig. 2. Held-out reconstruction quality and physical retransmission rate; bands show 95% confidence intervals across independent run pairs.

B. Post-Flow HARQ Decision

A quality head predicts log-MSE from the repaired tokens, unreliability scores, and receiver CSI, then converts it to predicted PSNR $\hat{q}$. FlowHARQ ACKs when $\hat{q} \geq q_0$ and otherwise decodes (3). The DeepJSCC transmitter is unchanged. The total training loss is

$$\mathcal{L} = \lambda_f \mathcal{L}_{\text{FM}} + \lambda_r \mathcal{L}_{\text{rel}} + \lambda_q \mathcal{L}_{\text{quality}} + \lambda_x \mathcal{L}_1. \quad (7)$$

IV. EXPERIMENTAL PROTOCOL

We initialize the encoder/decoder from a fixed SwinJSCC checkpoint and train only RAFM, reliability, and quality heads on random 256×256 DIV2K crops [10]. Training samples SNR in $[0, 15]$ dB, speed in $[0, 120]$ km/h, and CSI age in $[0, 5]$ ms. Evaluation uses SNR $\{0, 3, 6, 9, 12, 15\}$ dB, speeds $\{30, 60, 90, 120\}$ km/h, a 1 ms HARQ interval, and Rayleigh fading. The first 20 validation images form a calibration split; the remaining 80 are never used for model, mask, ODE-step, threshold, or bias selection.

Calibration chooses $K \in \{1, 2, 4, 8\}$ and $\tau_r \in \{0, .5, .65, .8, .9\}$, then fits one scalar bias for each quality head. The frozen configuration is $K = \text{TBD}$ and $\tau_r = \text{TBD}$. We compare direct DeepJSCC, full two-round HARQ, FM without HARQ, adaptive HARQ, FlowHARQ, and oracle variants whose NACK uses ground-truth PSNR. Every method shares the same first- and second-round channel draws. We report PSNR, SSIM [11], LPIPS [12], physical retransmission rate, mean rounds, and GPU latency. Confidence intervals are two-sided 95% Student- t intervals over independent training-seed means, after averaging channel seeds; paired effects are FlowHARQ minus adaptive HARQ within each training seed.

V. RESULTS

At the frozen calibration point, FlowHARQ changes the mean physical retransmission rate by TBD percentage points relative to adaptive HARQ, with a paired 95% CI of [TBD, TBD]. The corresponding PSNR change is TBD dB ([TBD, TBD]), and the LPIPS change is TBD. Figure 2 separates these effects by SNR.

The receiver-side virtual round takes TBD ms per image on an RTX PRO 6000, compared with TBD ms for local second-round combining and decoding; over-the-air scheduling and propagation are excluded. This latency result therefore measures receiver computation, not complete network latency.

VI. LIMITATIONS AND CONCLUSION

FlowHARQ currently uses a flat-fading simulation, a fixed DeepJSCC backbone, and full rather than token-selective physical retransmission. The quality head requires a small held-out calibration split, and receiver compute is not free. Within this controlled scope, the experiments test whether partial latent FM can substitute for some physical HARQ rounds. The central claim is supported only if the retransmission confidence interval excludes zero while PSNR and LPIPS remain within the predeclared matched-quality tolerances; otherwise the result should be reported as a negative or inconclusive finding.

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